INCH-POUND

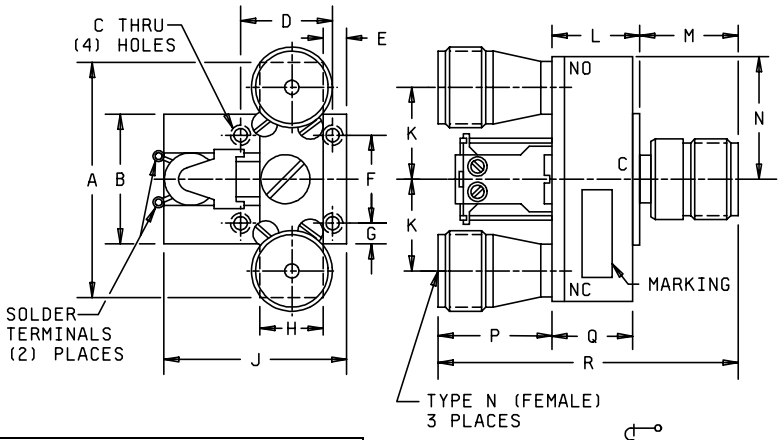
MIL-DTL-3928/10E
15 August 2001
SUPERSEDING
MIL-S-3928/10D
15 July 1988

DETAIL SPECIFICATION SHEET

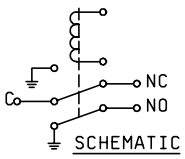
SWITCHES, RADIO-FREQUENCY TRANSMISSION LINE
(COAXIAL) (ELECTRICALLY OPERATED) CLASS 1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein shall consist of this specification and MIL-DTL-3928.



Position 1. (De-energized or failsafe position), C connects to NC connector.
Position 2. (Energized position), C connects to NO connector.



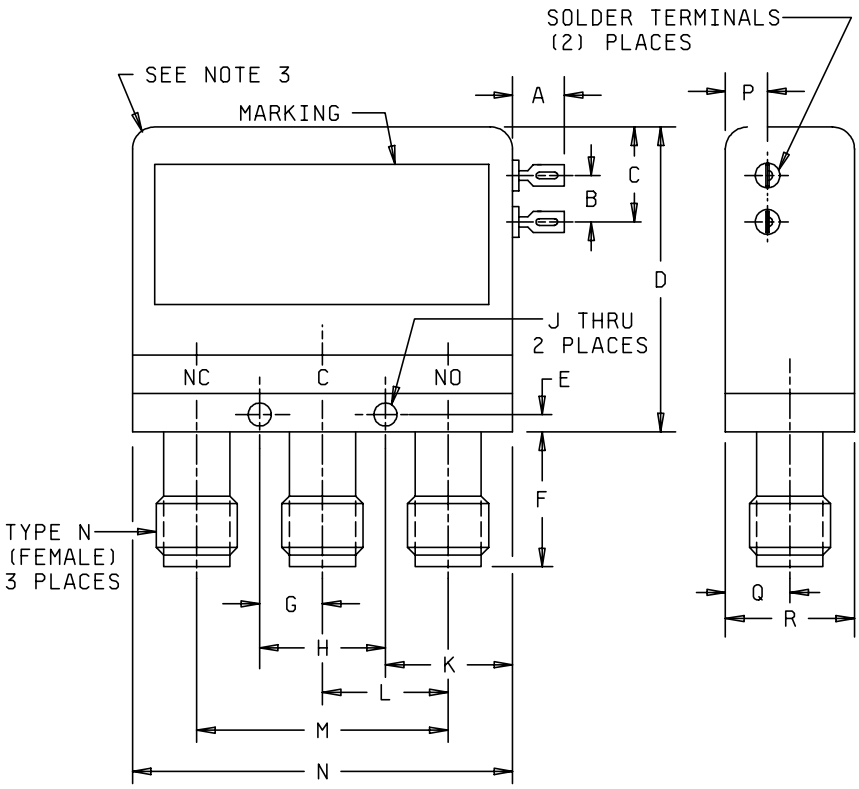
Dimensions									
	Inches		Millimeters			Inches		Millimeters	
Letter	Max	Min	Max	Min	Letter	Max	Min	Max	Min
A	2.03	1.97	51.6	50.0	J	1.510	1.490	38.35	37.85
B	1.135	1.115	28.83	28.32	K	.792	.772	20.12	19.61
C	.112-40 UNC-2B dia		2.84-40 UNC-2B dia		L	.604	.584	15.34	14.83
D	.697	.677	17.70	17.20	M	.681	.661	17.30	16.79
E	.104	.084	2.64	2.13	N	1.03	.97	26.2	24.6
F	.885	.865	22.48	21.97	P	.775	.755	19.68	19.18
G	.135	.115	3.43	2.92	Q	.510	.490	12.95	12.45
H	.510	.490	12.95	12.45					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 1. PINs M3928/10-01, M3928/10-02, and M3928/10-03.

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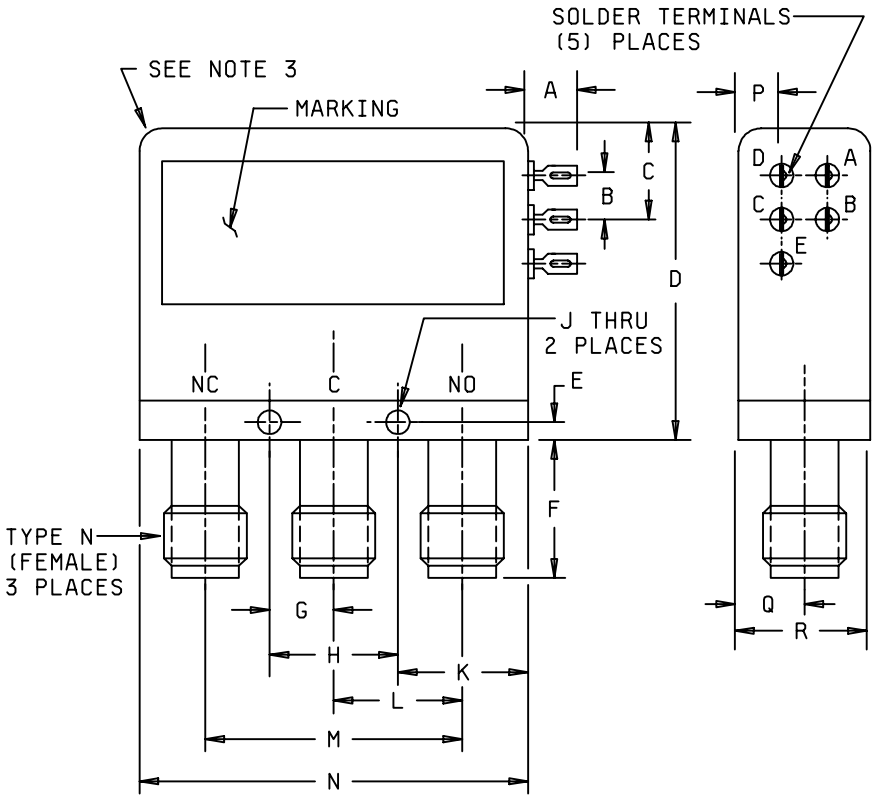
Position 1. (De-energized or failsafe position),
C connects to NC connector.
Position 2. (Energized position),
C connects to NO connector.

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.32	---	8.1	---	J	.213 dia	.193 dia	5.41 dia	4.90 dia
B	.38	.32	9.7	8.1	K	.880	.860	22.35	21.84
C	.73	.67	18.5	17.0	L	.895	.875	22.73	22.23
D	2.2	---	56	---	M	1.780	1.760	45.21	44.70
E	.210	.190	5.33	4.83	N	2.71	---	68.8	---
F	.63	.57	16.0	14.5	P	.27	.21	6.9	5.3
G	.452	.432	11.48	10.97	Q	.56	.50	14.2	12.7
H	.895	.875	22.73	22.23	R	1.09	---	27.7	---

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 2. PINs M3928/10-04 and M3928/10-09.

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Position 1. (De-energized or failsafe position), C connects to NC connector.
Position 2. (Energized position), C connects to NO connector.

Actuator and ind. ckt terminals	
Solenoid common	A
Solenoid (position 2) +28 V dc	B
Ind. ckt common	C
Ind. ckt (position 1)	D
Ind. ckt (position 2)	E

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.32	---	8.1	---	J	.213 dia	.193 dia	5.41 dia	4.90 dia
B	.38	.32	9.7	8.1	K	.880	.860	22.35	21.84
C	.73	.67	18.5	17.0	L	.895	.875	22.73	22.23
D	2.2	---	56	---	M	1.780	1.760	45.21	44.70
E	.210	.190	5.33	4.83	N	2.71	---	68.8	---
F	.63	.57	16.0	14.5	P	.27	.21	6.9	5.3
G	.452	.432	11.48	10.97	Q	.555	.535	14.10	13.59
H	.895	.875	22.73	22.23	R	1.09	---	27.7	---

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 3. PIN M3928/10-05.

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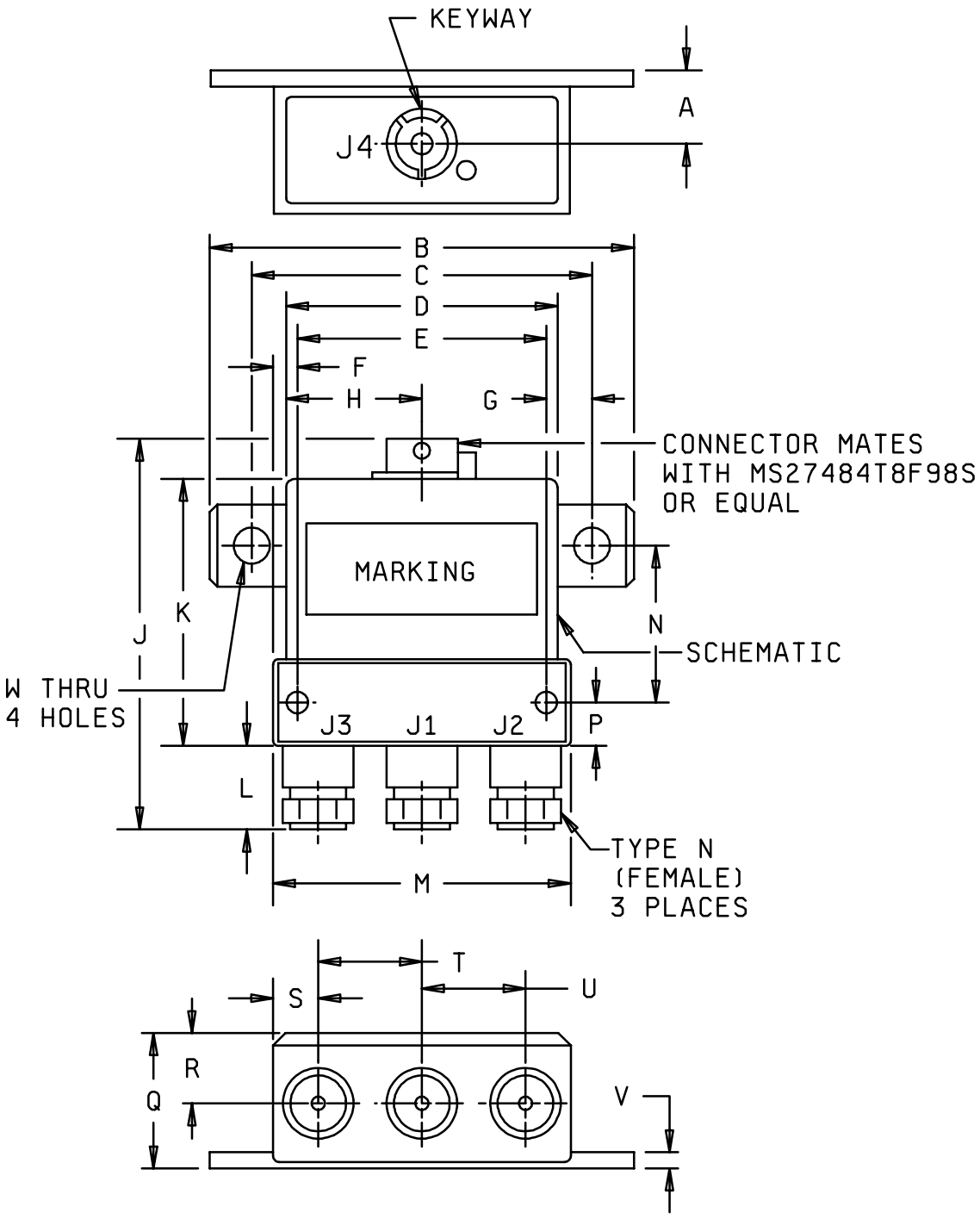
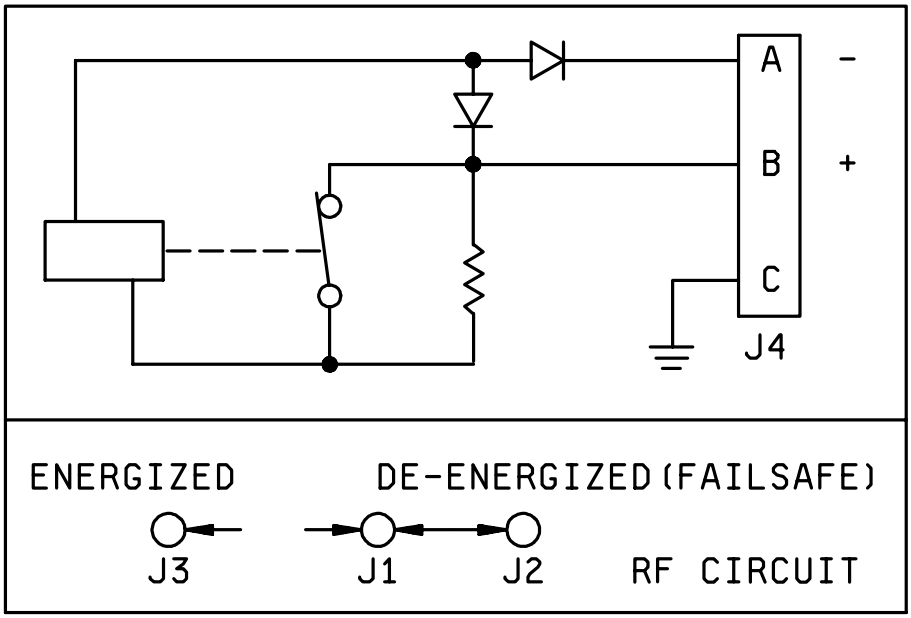


FIGURE 4. PIN M3928/10-06.

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SCHEMATIC



Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.59	.53	15.0	13.5	M	2.72	2.66	69.1	67.6
B	3.47	3.41	88.1	86.6	N	1.545	1.535	39.2	39.0
C	2.945	2.935	74.80	74.55	P	.39	.33	9.9	8.4
D	2.53	2.47	64.3	62.7	Q	1.15	1.09	29.2	27.7
E	2.395	2.385	60.83	60.58	R	.59	.53	15.0	13.5
F	.18	.12	4.6	3.0	S	.50	.44	12.7	11.2
G	.30	.24	7.6	6.1	T	.873	.867	22.17	22.02
H	1.28	1.22	32.5	31.0	U	.873	.867	22.17	22.02
J	3.66	---	93.0	---	V	.10	.04	2.5	1.0
K	2.63	---	66.8	---	W	.185	.175	4.70	4.44
L	.69	.63	17.5	16.0					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 4. PIN M3928/10-06 - Continued.

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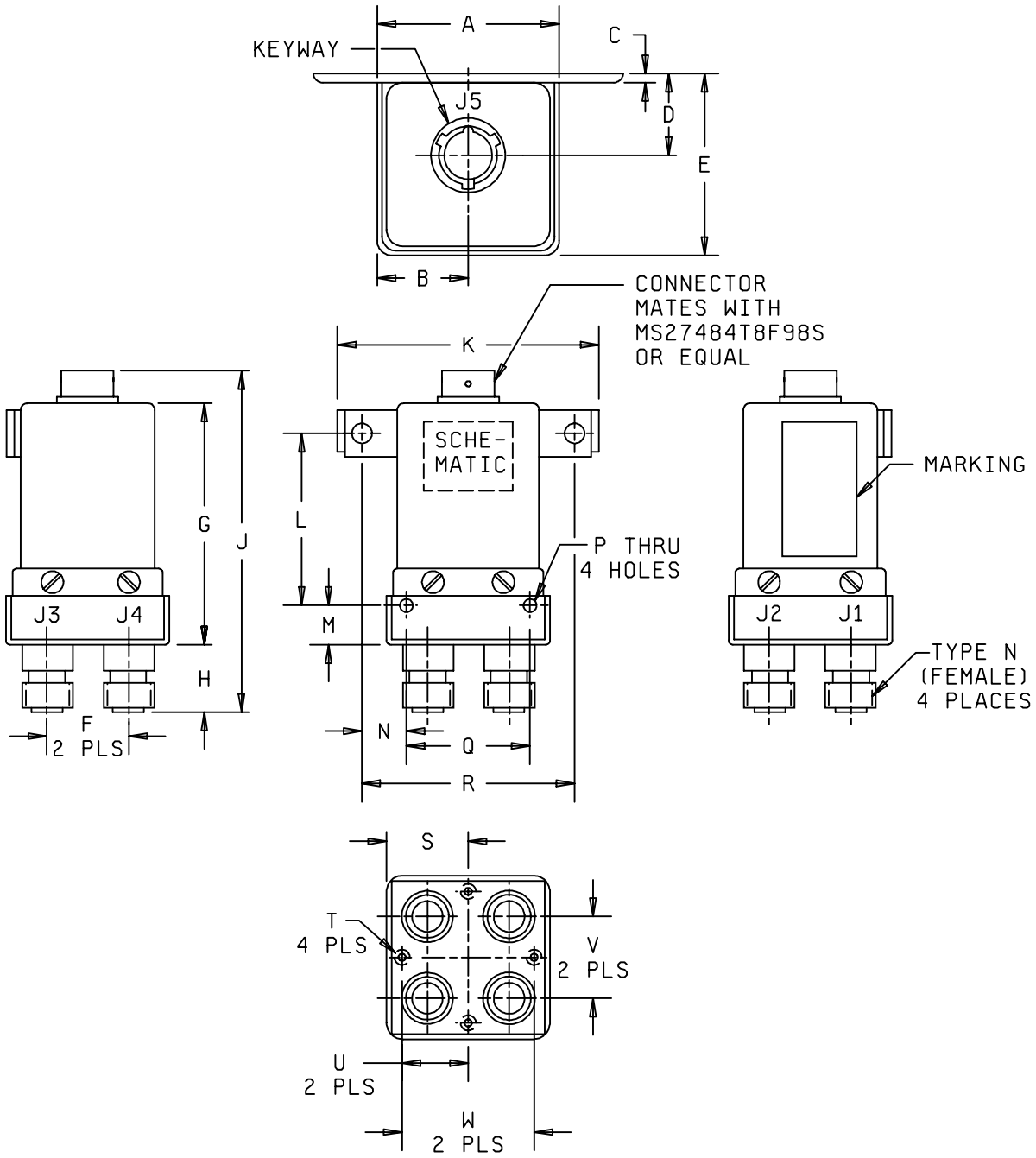
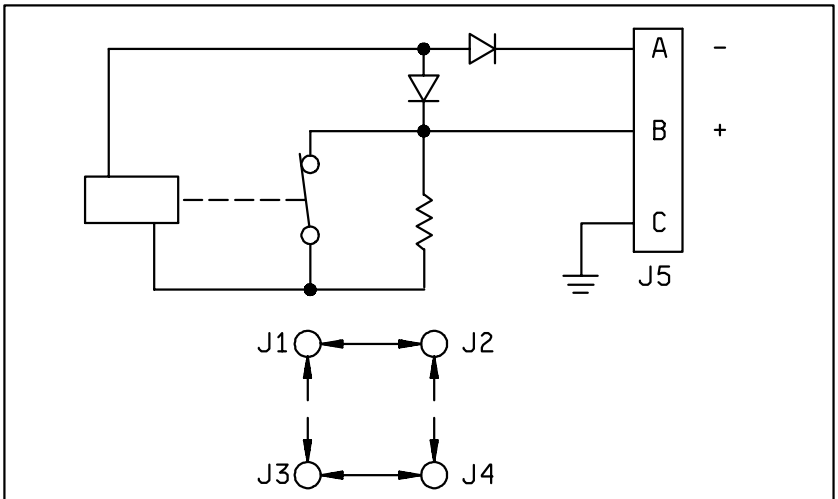


FIGURE 5. PIN M3928/10-07.

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SCHEMATIC



POSITION 1 (FAILSAFE POSITION) J1 - J2
J3 - J4

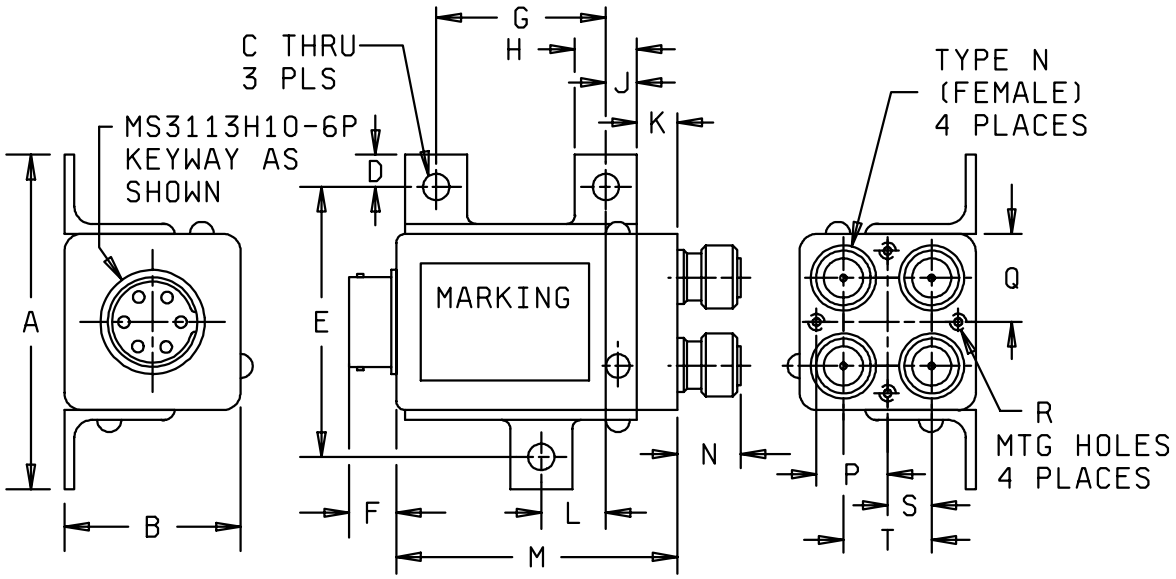
POSITION 2 (ENERGIZED POSITION) J1 - J3
J2 - J4

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.00	---	50.8	---	M	.40	.33	10.2	8.4
B	1.10	.90	27.9	22.9	N	.37	.31	9.4	7.9
C	.107	.097	2.72	2.46	P	.185 dia	.175 dia	4.70 dia	4.45 dia
D	1.10	.90	27.9	22.9	Q	1.567	1.557	39.80	39.55
E	2.00	---	50.8	---	R	2.255	2.245	57.28	57.02
F	1.03	.97	26.2	24.6	S	1.010 ref	.990 ref	25.65 ref	25.15 ref
G	3.28	3.22	83.3	81.8	T	.164-32 UNC-2B x .37 DP	(No. 8)	4.17-32 UNC-2B x 9.40 DP	(No. 8)
H	.69	.63	17.5	16.0					
J	4.40	---	111.8	---	U	.817	.807	20.75	20.50
K	2.78	2.72	70.6	69.1	V	1.03	.97	26.2	24.6
L	2.40	2.34	61.0	59.4	W	1.630	1.620	41.40	41.15

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 5. PIN M3928/10-07 - Continued.

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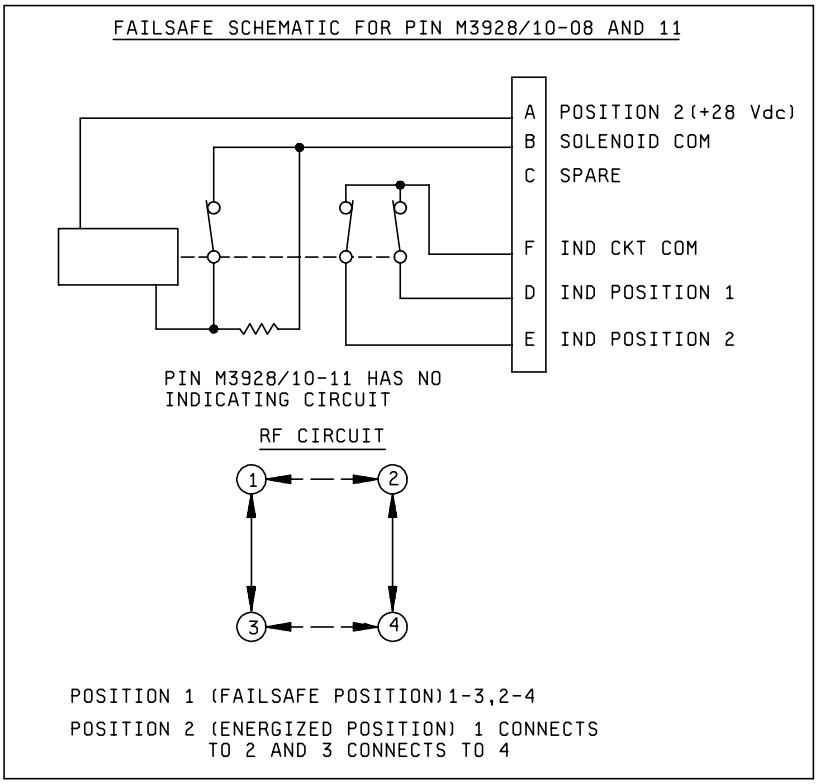
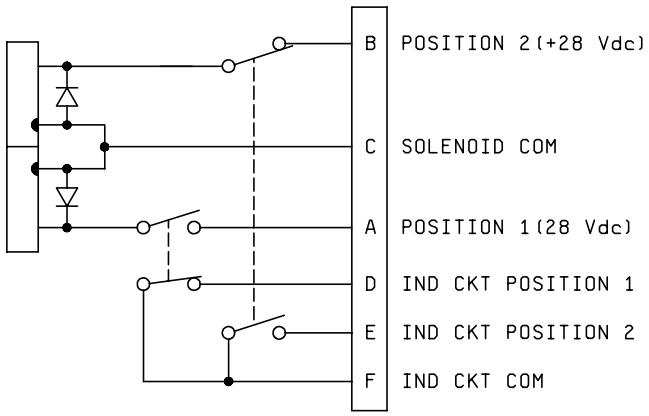
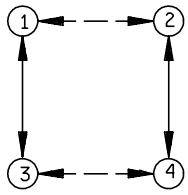


FIGURE 6. PINs M3928/10-08, M3928/10-10, and M3928/10-11.

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LATCHING SCHEMATIC FOR PIN M3928/10-10





POSITION 1 ⟷ 1-3,2-4
POSITION 2 ⟷ 1-2,3-4

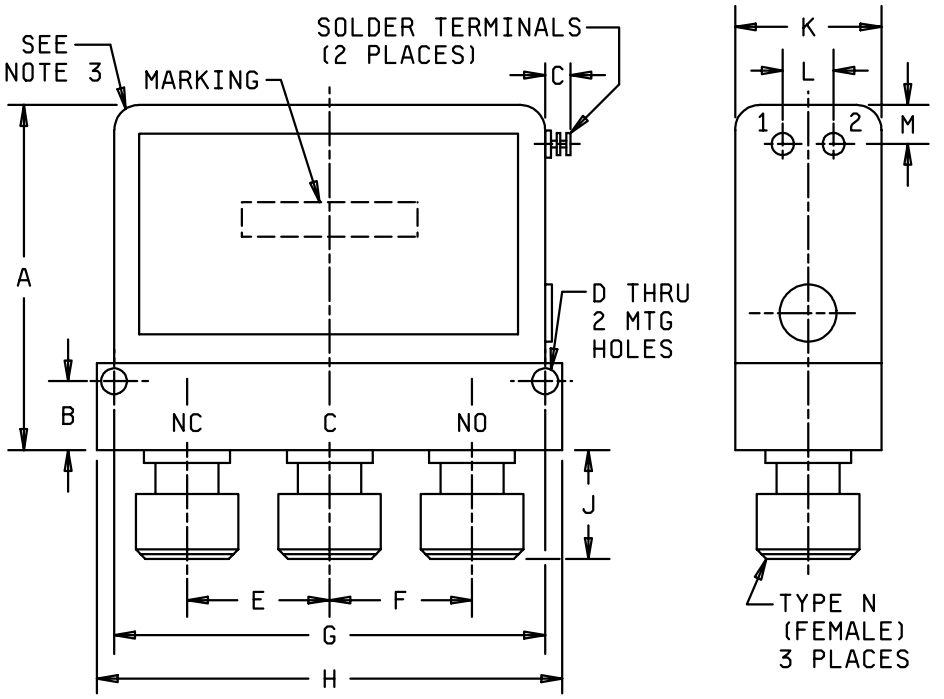
RF CIRCUIT

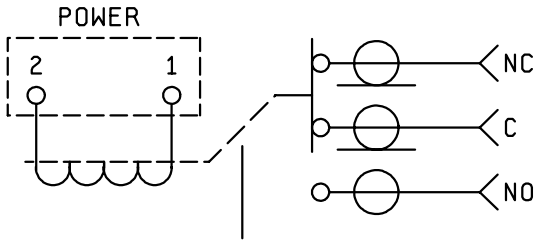
Dimensions									
	Inches		Millimeters			Inches		Millimeters	
Letter	Max	Min	Max	Min	Letter	Max	Min	Max	Min
A	3.28	3.22	83.3	81.8	L	.753	.747	19.13	18.97
B	1.78	---	45.2	---	M	2.60	---	66.0	---
C	.198 dia	.192 dia	5.03 dia	4.88 dia	N	.65	---	16.5	---
D	.35	.29	8.9	7.4	P	.730	.710	18.54	18.03
E	2.610	2.590	66.29	65.79	Q	.885	.865	22.48	21.97
F	.49	---	12.4	---	R	.138-32 UNC-2B		3.50-32 UNC-2B	
G	1.510	1.490	38.35	37.85		x .25 DP	(No. 6)	x 6.4 DP	(No. 6)
H	.63	.57	16.0	14.5	S	.452	.432	11.48	10.97
J	.33	.27	8.4	6.9	T	.895	.875	22.73	22.23
K	.28	.22	7.1	5.6					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 6. PINs M3928/10-08, M3928/10-10, and M3928/10-11 - Continued.

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Position 1. (De-energized or failsafe position), C connects to NC connector.
Position 2. (Energized position), C connects to NO connector.

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	1.80	---	45.7	---	G	2.53	2.47	64.3	62.7
B	.390	.370	9.91	9.40	H	2.78	2.72	70.6	69.1
C	.32	---	8.1	---	J	.63	.57	16.0	14.5
D	.166	.156	4.22	3.96	K	.93	.87	23.6	22.1
E	.890	.870	22.61	22.10	L	.28	.22	7.1	5.6
F	.890	.870	22.61	22.10	M	.23	.17	5.8	4.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 7. PIN M3928/10-12.

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TABLE I. Electrical and performance characteristics.

PIN M3928/10-	Fig. No.	Housing	Config.	Manual or remote	Solenoid or motor	Fail-safe or latching	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switching time (ms)	Position indicating circuit	Life cycles x 1000	Operating voltage	Pick-up voltage	Drop-out voltage	Operating current (A) 1/	Holding current fail-safe type 1/ (A)	Power and indicator connector	Weight (oz) 2/
								max	max	min	max			nom	max	min	max			max
01N, S	1	O	1P2T	R	S	F	5.2	1.4:1	0.48	25	20	None	1000	110 V dc	84 V dc	10 V dc	0.2	0.2	Solder terminals	8 (227)
02N, S 3/	1	O	1P2T	R	S	F	5.2	1.4:1	0.48	25	20	None	1000	28 V dc	18 V dc	3 V dc	0.2	0.2	Solder terminals	8 (227)
03N, S 3/	1	O	1P2T	R	S	F	5.2	1.4:1	0.48	25	20	None	1000	115 V ac	93 V ac	5 V ac	0.2	0.2	Solder terminals	8 (227)
04N, S	2	E	1P2T	R	S	F	12.4	1.5:1	0.5	60	50	None	100	28 V dc	18 V dc	3 V dc	0.8	0.4	Solder terminals	8 (227)
05N, S	3	E	1P2T	R	S	F	12.4	1.5:1	0.5	60	50	Yes	100	28 V dc	18 V dc	3 V dc	0.8	0.4	Solder terminals	8 (227)
06N, S	4	H	1P2T	R	S	F	12.4	1.5:1	0.5	25	20	None	50	28 V dc	20 V dc	14 V dc	1.0	0.5	N/A	12 (342)
07N, S	5	H	TR	R	S	F	12.4	1.5:1 5/	0.5 6/	25 7/	30 4/ 20	None	50	28 V dc	20 V dc	14 V dc	1.5	0.6	N/A	16 (456)
08N, S	6	H	TR	R	S	F	12.4	1.5:1 5/	0.5 6/	60 7/	30 4/ 30	Yes	100	28 V dc	20 V dc	14 V dc	1.0	0.3	N/A	16 (456)
09N, S	2	E	1P2T	R	S	F	12.4	1.5:1 5/	0.5 6/	60 7/	50	None	100	115 V dc	95 V ac	5 V ac	0.12	.12	Solder terminals	8 (227)
10N, S	6	H	TR	R	S	L	12.4	1.5:1 5/	0.5 6/	60	20	Yes	100	28 V dc	20 V dc	---	0.64	---	N/A	12 (342)
11N,S	6	H	TR	R	S	F	12.4	1.5:1	0.5	60	20	None	100	28 V dc	20 V dc	14 V dc	1.5	.25	N/A	12 (342)
12N, S	7	E	1P2T	R	S	F	.96 to 1.215	1.2:1	0.3	60	30	None	100	28 V dc	20 V dc	14 V dc	0.2	0.2	Solder terminals	8 (227)

1/ At nominal operating voltage and 20°C.

2/ Mass in grams in parentheses.

3/ Open port is grounded.

4/ At low and high temperature extremes.

5/ The VSWR at .225-.400 GHz for dash numbers 06 and 07 is 1.1:1. The VSWR at .100-.500 GHz for dash number 08 is 1.15:1. The VSWR at 4 GHz for dash number 10 is 1.25:1.

6/ The insertion loss at .225-.400 GHz for dash numbers 06 and 07 is 0.1 dB. The insertion loss at .100-.500 GHz for dash number 08 is 0.15 dB. The insertion loss at 4 GHz for dash number 10 is 0.25 dB.

7/ The isolation at .225-.400 GHz for dash numbers 06 and 07 is 50 dB. The isolation at .100-.500 GHz for dash number 08 is 80 dB.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 7.

RF connectors: Female connectors shall mate with type N connectors in accordance with MIL-PRF-39012/1.

Nominal impedance: 50 ohms.

Termination: Open.

RF power handling capability (average): 100 watts for dash numbers 01 through 11 and 280 watts for dash number 12.

RF connectors: Series N connectors shall mate with series N male connectors in accordance with MIL-PRF-39012/1.

Electrical and performance characteristics: See table I.

Vibration: Method I.

Operating temperature: -55 deg C to +85 deg C.

Part or Identifying Number (PIN): M3928/10- (dash number from table I).

TABLE II. PIN to type cross-reference.

PIN M3928/10-	Type SA- /U
01	1360
02	1361
03	1362
04	---
05	---
06	---
07	---
08	---
09	---
10	---
11	---
12	---

Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5985-1209-04)

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